

Rapid Screening of point mutation by mismatch amplification mutation assay PCR

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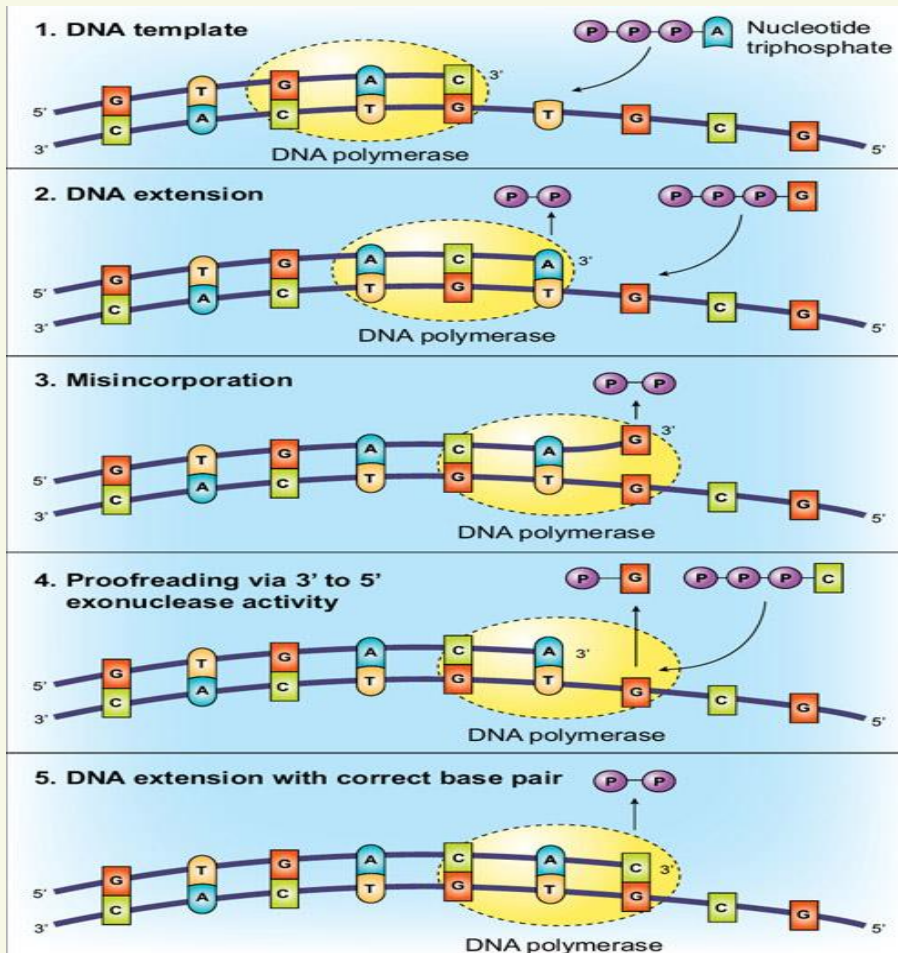




WHAT IS PCR?

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=2KOLNIWOZKU](https://www.youtube.com/watch?v=2KOLNIWOZKU)

THERMOSTABLE POLYMERASES USED FOR PCR



- Most second-generation thermostable polymerases have 3' to 5' exonuclease activity.
- Exonuclease activity contributes to their high fidelity, enabling them to remove misincorporated bases.
- Taq polymerase lacks this exonuclease activity, which is important for MAMA PCR.
- If the primer & DNA are not perfectly complexed, then Taq polymerase is not able to extend the DNA from the primer.

What does exonuclease activity mean for a DNA polymerase?

<https://www.neb.com/en-us/faqs/what-does-exonuclease-activity-mean-for-a-dna-polymerase?srsId=AfmBOopyK0vkXR3xtK-bl2gYTQBJdur5OIh3AttkP09QX2eF5S35-nq>

https://www.neb.com/en-us/-/media/nebus/files/feature-articles/images/fa_polymerase_figure2.jpg?rev=2722cdd8233e4d4ab828ada9a543a04f&hash=6016229CE231BECCF3D1A4A102F0F6DE&device=modal

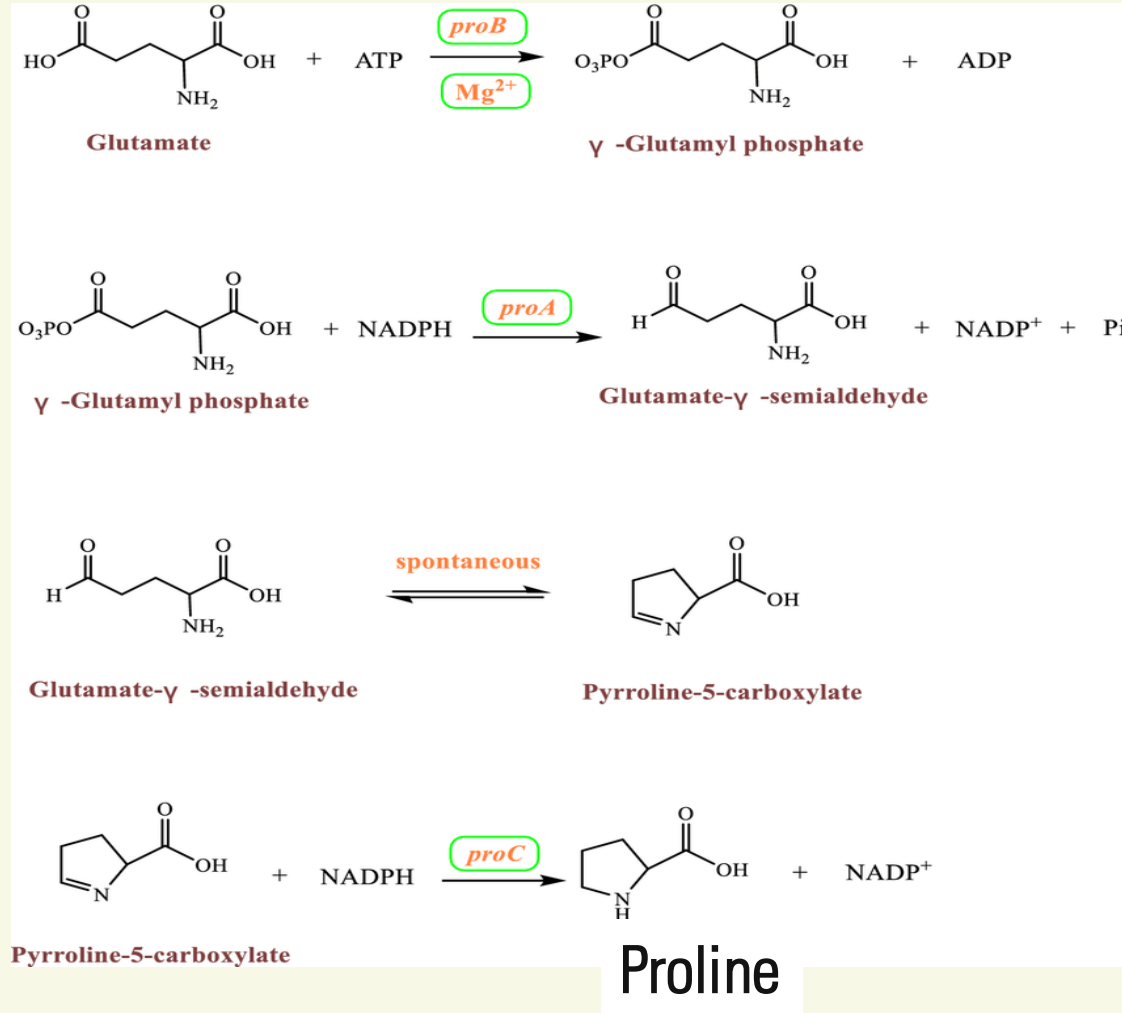
Why choosing MAMA PCR Technique

Background and Motivation

GOAL OF STUDY



succinate $\leftarrow$ 2-oxoglutarate $\rightarrow$ glutamate



- Corynebacterium glutamicum* is used for agricultural purposes to synthesize amino acids
- Goal this study was to screen through mutants to identify a strain with weakened 2-oxoglutarate dehydrogenase E1 subunit activity (*odhA*) weakening the conversion into succinate
- Increased glutamate arise from 2-oxoglutarate back conversion that can lead to increased proline production



Why Point Mutation Screening

Hard to detect by standard PCR

Single-nucleotide substitutions keep amplicon length unchanged, so gel-based screening can't differentiate mutants from wild-type sequences.

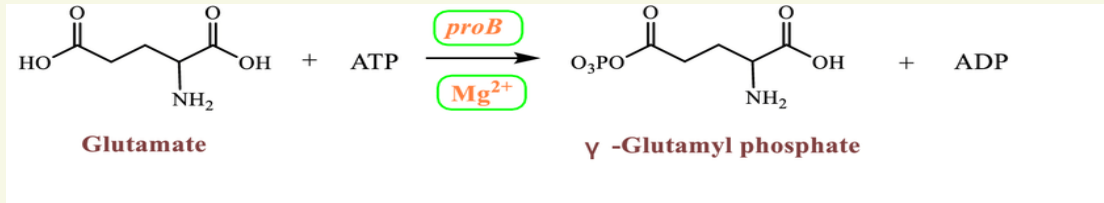
Low-efficiency edits inflate workload

Revertants (recombination) and CRISPR "escape" survivors can dominate, forcing sequencing of many colonies to find a few true mutants.

Sequencing slows design-build-test

Outsourced turnaround adds days to weeks and raises costs—limiting iterative workflows like RBS tuning, base editing, anpre-screend directed evolution. This motivates a fast before sequencing (optimized MAMA-PCR).

MAMA-PCR Methodology and Optimization



ProB mutated to proB^{G149K} relieve proline feedback inhibition

Study Goal: Robust MAMA-PCR Screening

Problem: selective amplification is fragile

Classical MAMA-PCR leverages Taq's mismatch sensitivity at the primer 3' end, but performance depends heavily on primer design, annealing temperature, and sequence context.

Approach: tune the key variables

Systematically optimize 3' mismatch count, primer melting-temperature pairing, and cycling annealing temperature to enable confident single-nucleotide discrimination.

Validation: works in real engineering

Demonstrated across multiple mutations in *C. glutamicum* and applied to screen a CRISPR-Cas12a RBS library targeting *odhA* for improved L-proline production.

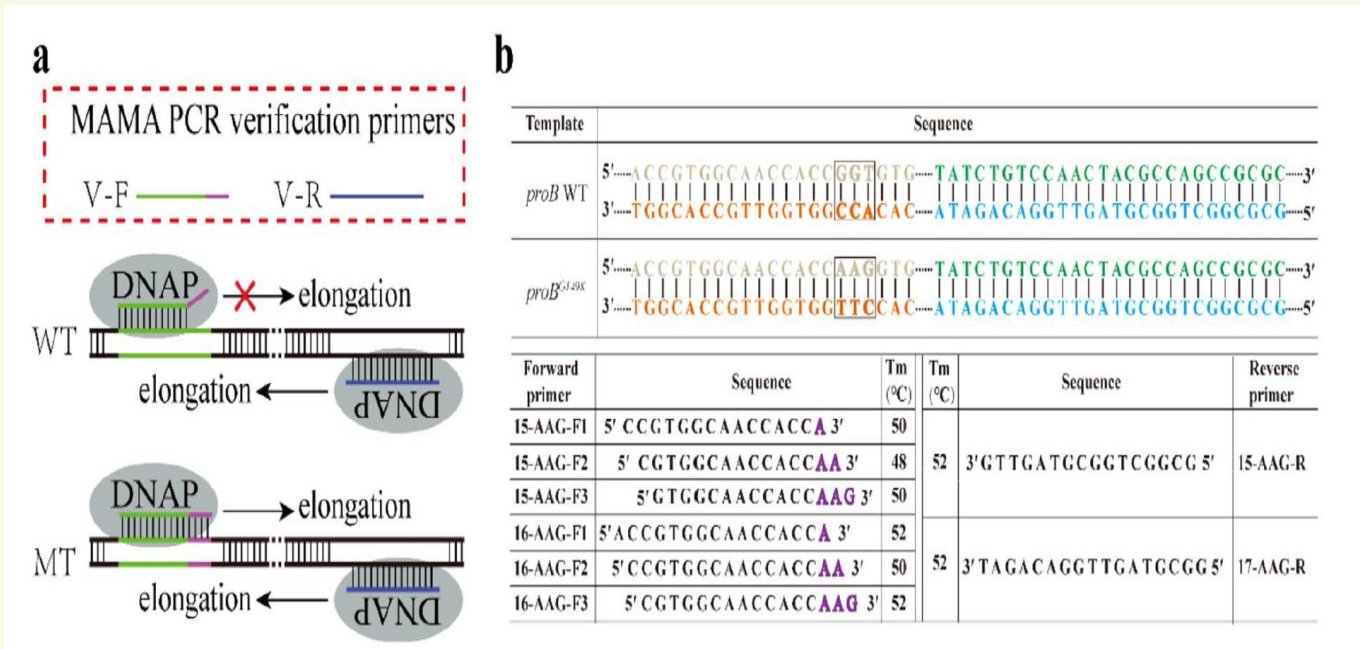


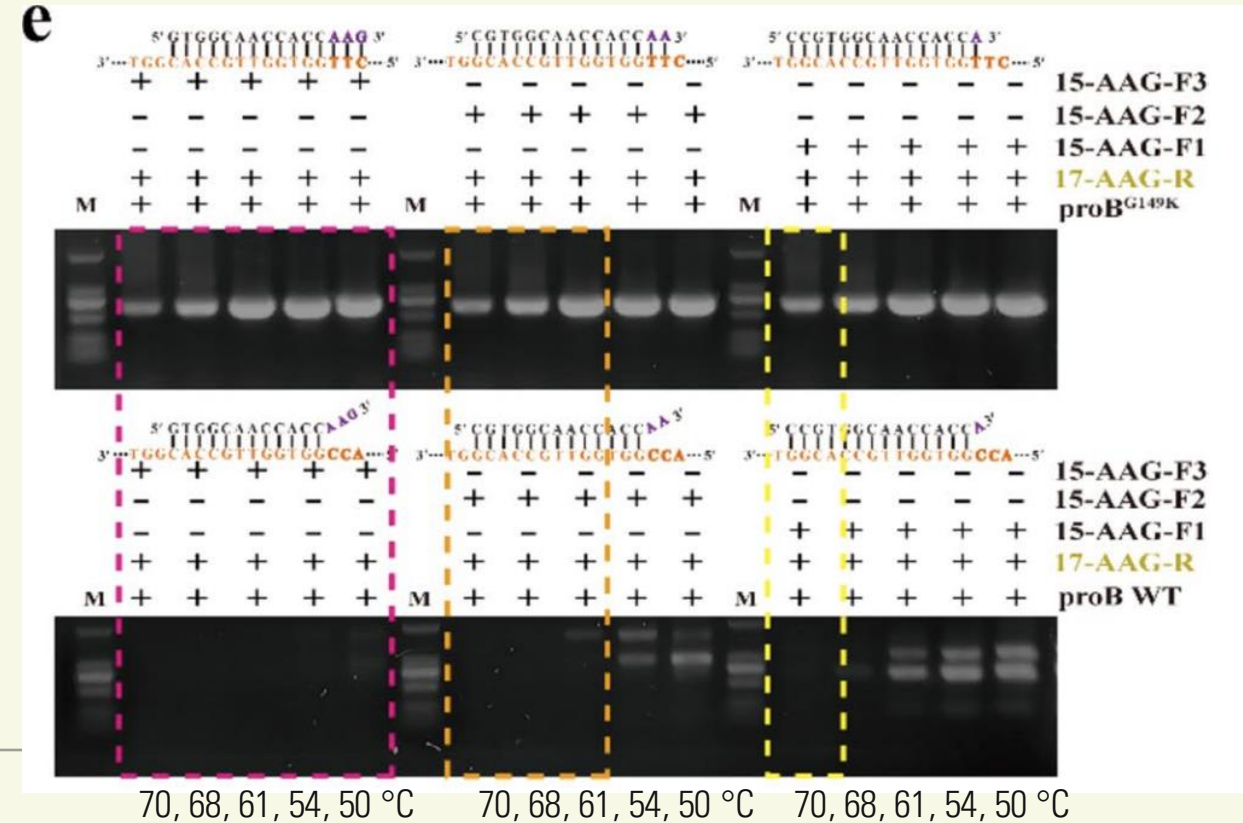
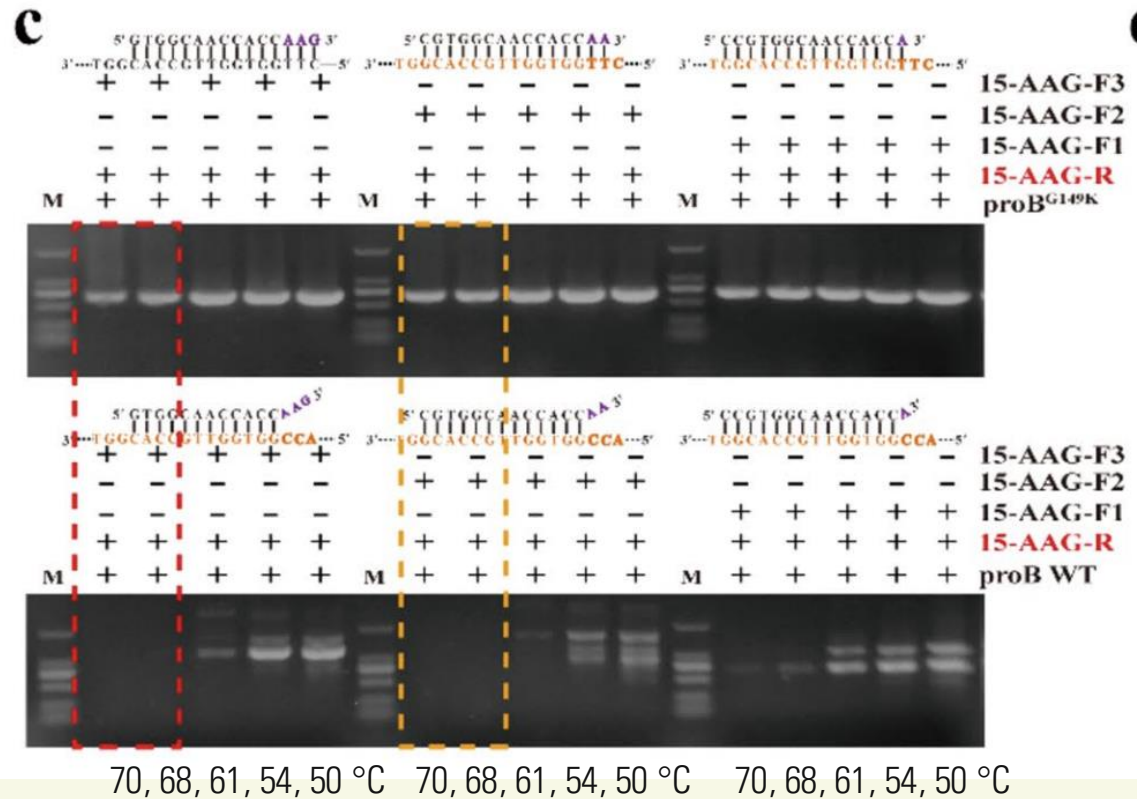
Fig. 1 Base substitutions at the *proB* site were identified using PCR.

PREAMPLIFICATION PCR IDENTIFIES PRIMERS ABLE TO DISTINGUISH MUTANT $\text{proB}^{\text{G149K}}$ FROM WT DNA

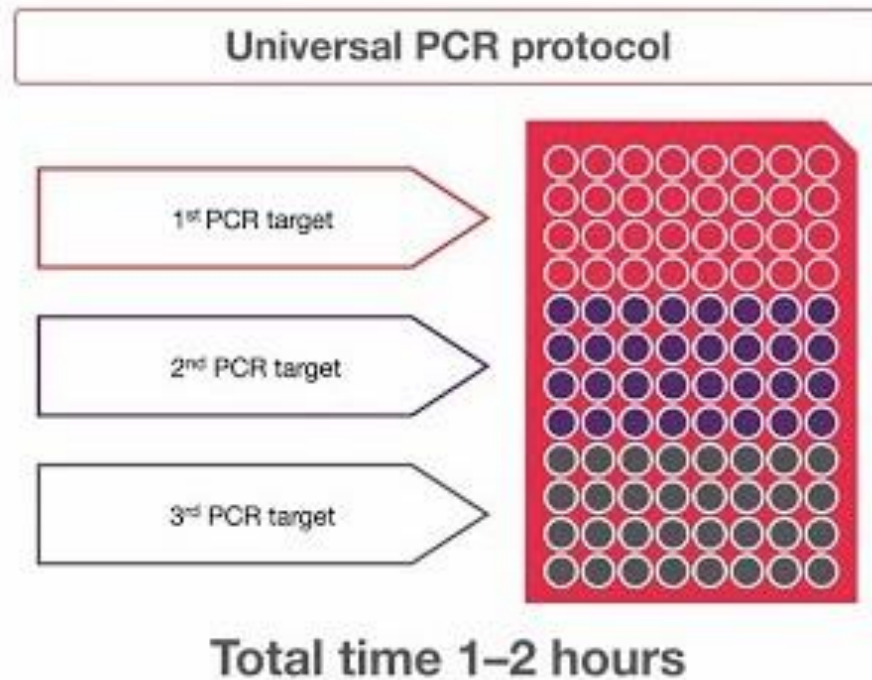
3 mismatched bases

2 mismatched bases

1 mismatched bases



Optimization Parameters in MAMA- PCR



3' mismatch number

Two to three mismatches near the 3' end sharply reduce wild-type extension while preserving mutant amplification.

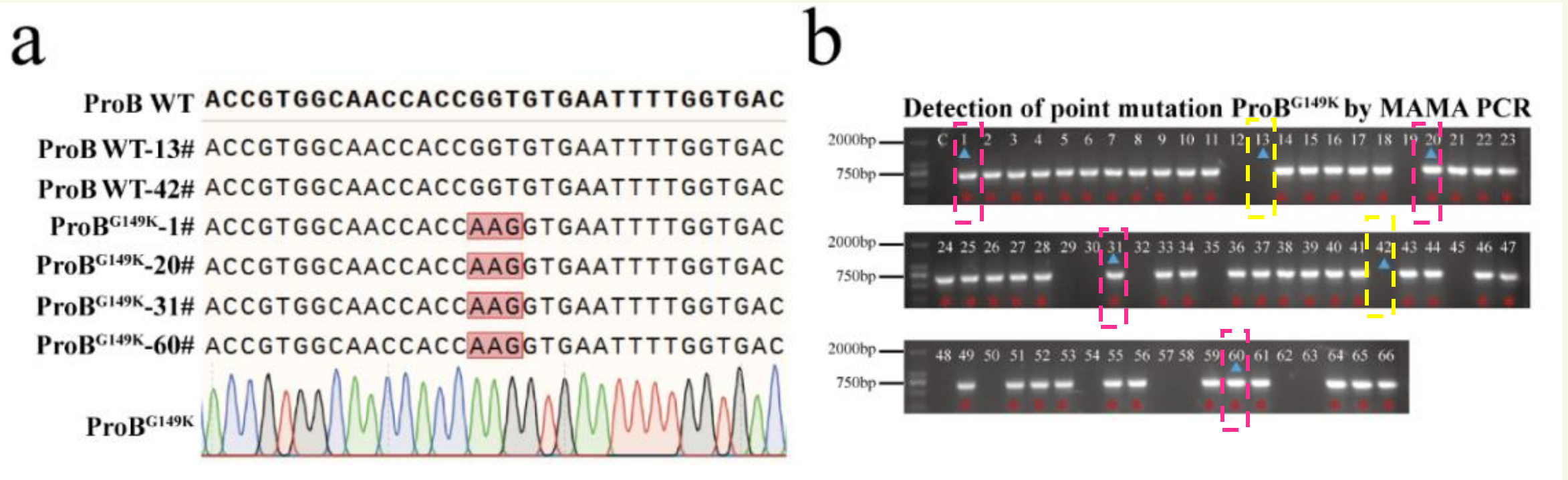
Primer pairing (T_m balance)

A higher- T_m partner primer increases dependence on correct 3' matching as annealing temperature rises.

Annealing temperature

Higher stringency magnifies mismatch effects; gradient PCR helps suppress wild-type signal without losing mutant detection.

SIX TRANSFORMANTS WERE RANDOMLY SELECTED FOR SEQUENCING



Results indicate that MAMA PCR correctly distinguishes mutation G149K in proB DNA from WT DNA

Mutating the Zwf and Gnd genes to improve NADPH availability

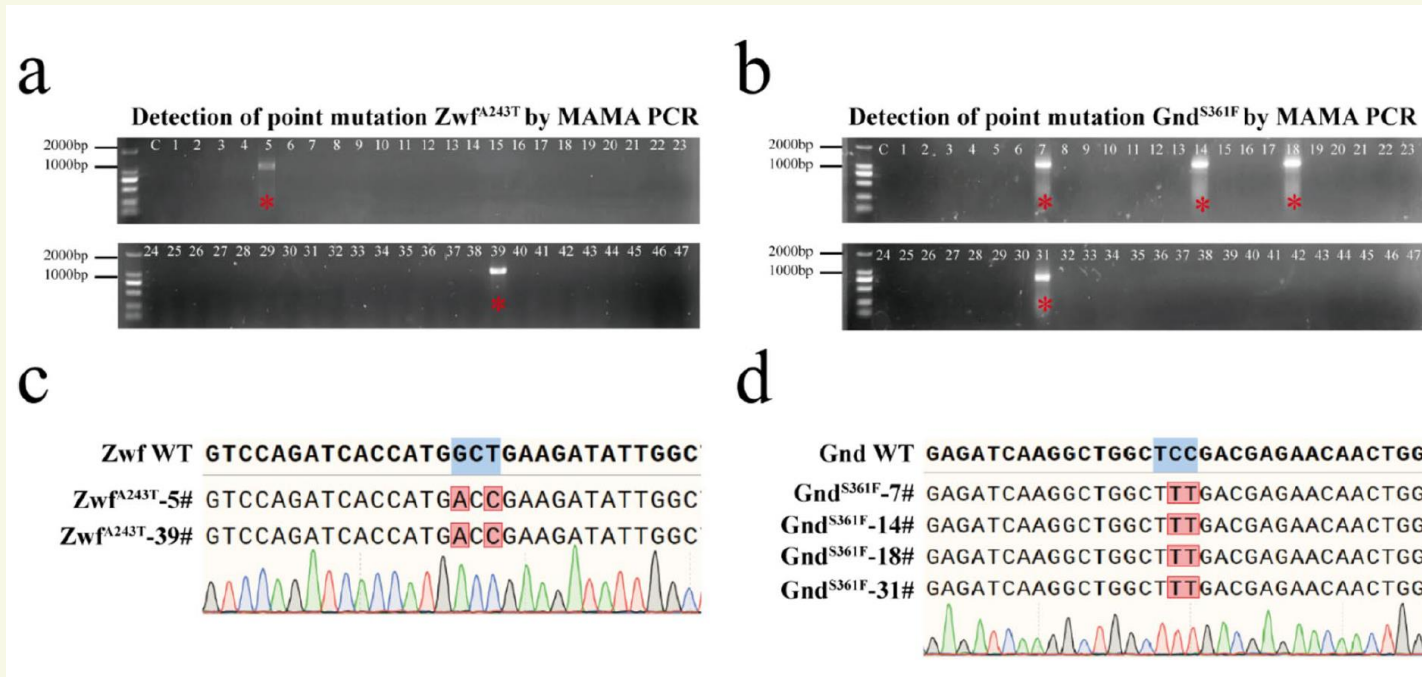


Fig. 4: Detection of *zwf* and *gnd* mutations by MAMA PCR

- c** Sequencing results of recombinants with identified mutations in **a**
- d** Sequencing results of recombinants with identified mutations in **b**

Controlled test system

Optimized MAMA-PCR is first applied to a known single-nucleotide proB mutation in *C. glutamicum*, enabling direct WT vs mutant comparison.

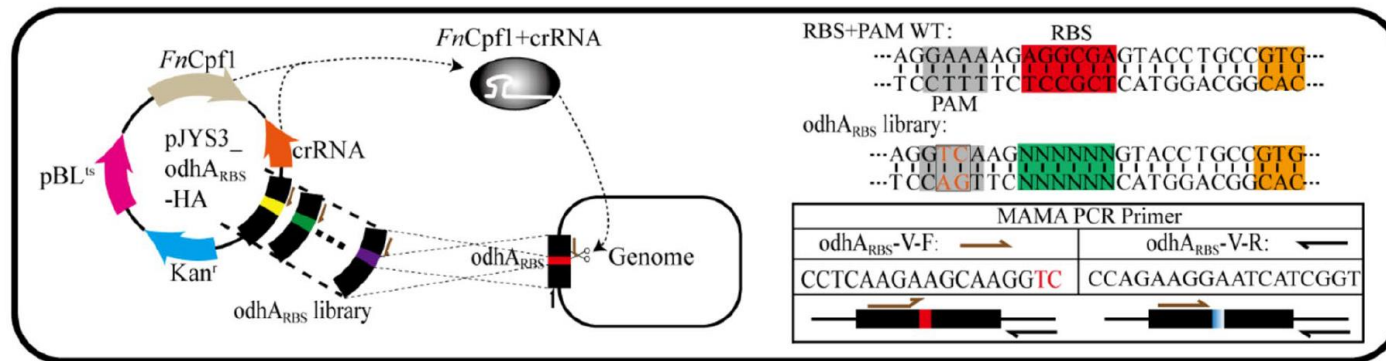
Clean allele discrimination

By tuning mismatch number and annealing temperature, amplification is robust only for the mutant template, with sequencing confirming the call.

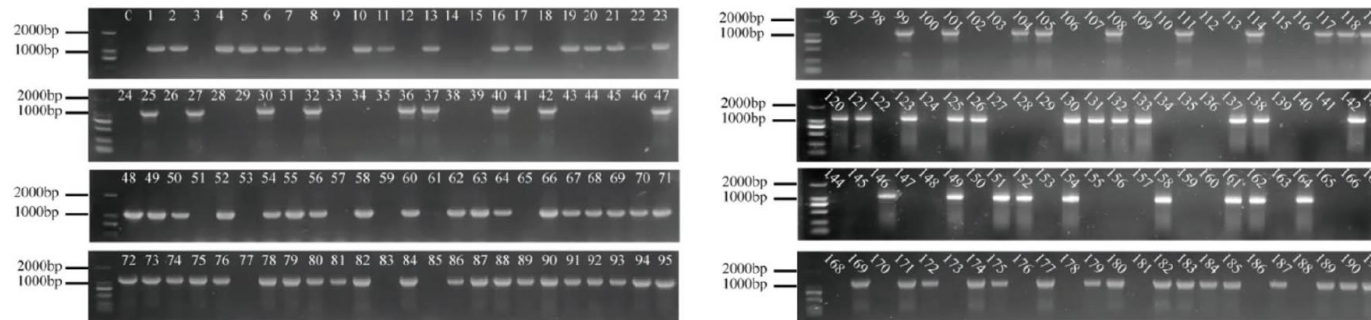
Why this validation matters

It isolates assay performance from confounders (low edit rates, mixed populations) and sets a benchmark before testing low-frequency mutants.

a



b

**Fig 2:**

a Application of the pJYS3_odhARBS-LIB plasmid librarybased CRISPR-FnCpf1 system for RBS engineering in front of the *odhA* gene. Two bases of AA at PAM were mutated to TC for destroying PAM and designing MAMA PCR primers. The 3' terminal of primer odhARBS- V-F cannot adhere to the wild-type template, but can effectively adhere to the mutant template. The primer odhARBS- V-R cannot be attached to the homologous arm.

b MAMA PCR verification results of mutant library. Transformants with bands indicate that the genome has mutated

MAMA-PCR and the 3' Mismatch Effect

Core principle

Taq polymerase poorly extends primers with a 3'-terminal mismatch, so a primer matching the mutant allele amplifies mutant templates while suppressing wild type.

Why false positives happen

Single mismatches can be tolerated—especially at lower annealing temperatures or in stabilizing sequence contexts—producing background amplification on wild-type templates.

Study framing

The mismatch effect is treated as tunable: specificity emerges from mismatch number, primer thermodynamics, and cycling conditions—supporting rational design and optimization.

Experimental Setup & PCR Conditions

Enzyme choice (why standard Taq)

Uses standard Taq (no 3'→5' exonuclease), enabling mismatch discrimination central to MAMA-PCR.

Replicable reaction + cycling details

Reports primer concentrations, template inputs, cycle numbers, and conditions sufficient for direct reproduction.

Gradient PCR + troubleshooting

Recommends gradient PCR to tune annealing temperature per primer set; warns low T_a can produce nonspecific bands.

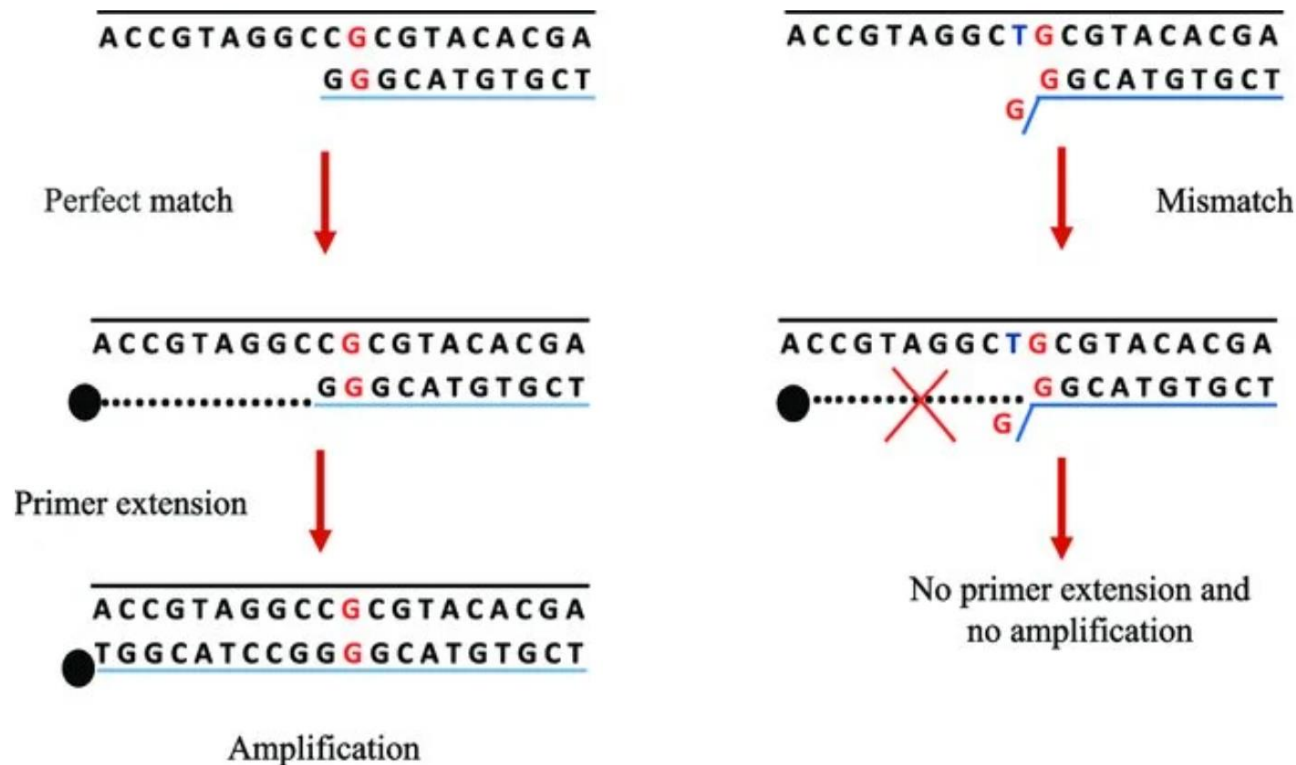


Fig. 3. Principle of the mismatch amplification mutation assay-polymerase chain reaction

Validation of the Optimized Method



Screening Low-Frequency Mutants

Rare-signal screening challenge

Optimized MAMA-PCR screens *zwf* A243T and *gnd* S361F recombinants amid many non-mutant revertants.

One-to-one validation

Only a small subset shows the expected band; sequencing confirms all positives are true mutants and all negatives are not.

Practical lab impact

Acts as a front-end filter, sharply reducing the number of samples requiring sequencing in real workflows.



Accuracy vs Sequencing

Validation agreement

Across experiments, MAMA-PCR matches DNA sequencing with no false positives or false negatives. Even small error rates could erode trust and offset time savings.

Workflow position

Sequencing remains the gold standard for confirmation. MAMA-PCR is framed as a rapid triage step that complements, not replaces, existing workflows.

Journal club prompts

What screening error rate is acceptable in practice? How well will these accuracy claims generalize across organisms, mutation types, and genomic contexts?

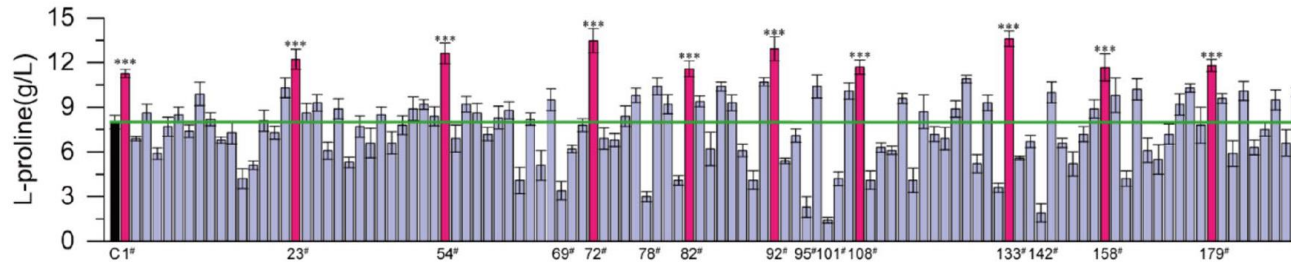
Application to Metabolic Engineering

58%

mutation efficiency from MAMA-PCR screening of an *odhA* RBS
mutant library

112/191 clones carried RBS mutations, eliminating 42% false positives early

c



The colonies were screened by 48-well plate fermentation and high-pressure liquid chromatography (HPLC) measurement of L-proline.

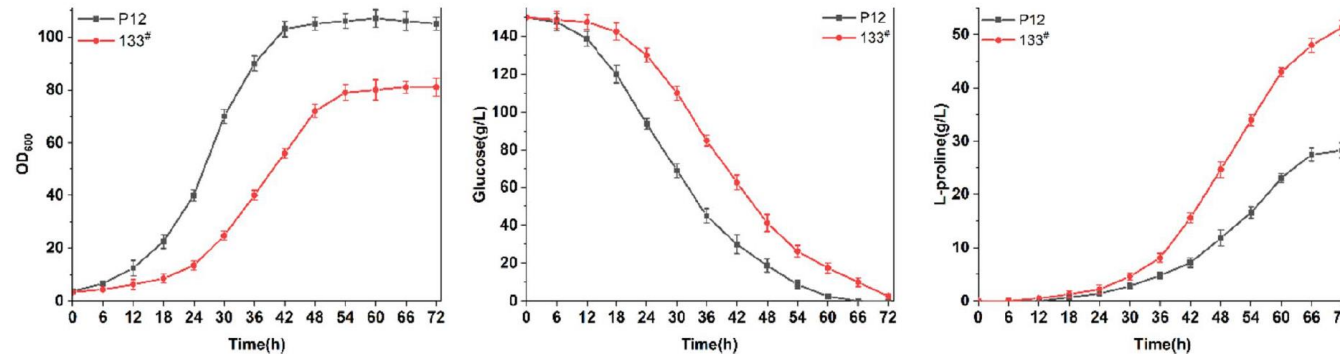


Fig5: Determination of fermentation performance of proline-producing strain 133# and P12 in shake flask.

a Growth curves of strains **133#** and P12

b glucose consumption of strains **133#** and P12

c Proline accumulation of strains **133#** and P12. Error bars indicate standard deviations from three parallel experiments

Functional Screening of Proline Producers

Why functional assays matter

After MAMA-PCR, selected mutants are tested for L-proline production to confirm that genotype changes translate into improved phenotype.

Key result: strain 133#

Strain 133# shows high proline titers despite slower growth and glucose consumption—highlighting a common flux redirection trade-off in metabolic engineering.

Takeaway for MAMA-PCR

Beyond genotyping, MAMA-PCR accelerates discovery by narrowing candidates before functional screening—enabling faster identification of superior producers.

51.3

g/L L-proline from mutant 133#

Shake-flask culture with no glutamate supplementation—matching parental performance with added glutamate.

Implications, Limitations, and Discussion

MAMA-PCR for Base-Editing Screens

Why screening limits base editing

Cytosine and adenine base editors make single-nucleotide changes without double-strand breaks, but often yield incomplete, mixed populations.

Where optimized MAMA-PCR fits

Rapid, low-cost screening identifies correctly edited clones, making base-editing workflows more practical in microbial systems.

Journal club discussion prompt

How could MAMA-PCR be adapted or multiplexed to keep pace with more complex, multi-site editing strategies?



Technical Limitations & Best Practices

Technical constraints

Requires Taq polymerase; not compatible with high-fidelity enzymes. Mismatch discrimination can be weaker at some sites, especially high-GC regions.

Primer design & optimization

Each new target needs empirical tuning up front, but effort is typically small compared with time and cost saved in large screening runs.

When to choose alternatives

Use MAMA-PCR for fast, targeted screening; consider amplicon sequencing when broader variant detection or higher confidence is required.

PCR Polymerases	3'→5' Exonuclease	Fidelity	5'→3' Exonuclease	Strand Displacement	Nick Translation	Extend RNA Primer	Extension from Nick	dU Tolerance	Resulting Ends	Popular Formats Available	Applications
High Fidelity PCR											
Q5® High-Fidelity DNA Polymerase	++++	280x <i>Taq</i>	No	—	No	No	No	No	Blunt	Q5® High-Fidelity DNA Polymerase	High-fidelity PCR, long range PCR, cloning from in vitro material for protein expression or gene analysis, SNP analysis via cloning and sequencing, site directed mutagenesis
										Q5® Hot Start High-Fidelity DNA Polymerase	
										Q5® High-Fidelity DNA Polymerase 2x Master Mix	
										Q5® High-Fidelity Hot Start DNA Polymerase 2x Master Mix	
Q5U® Hot Start High-Fidelity DNA Polymerase	++++		No	—	No	No	No	Yes	Blunt	Q5U® Hot Start High-Fidelity DNA Polymerase	USER cloning, high fidelity amplification of bisulfite converted or deaminated DNA substrates, carryover prevention
										NEBNext Q5U® Master Mix	
Phusion™ High-Fidelity DNA Polymerase*	++++	39 ^b -50x ^c <i>Taq</i>	No	—	No	No	No	No	Blunt	Phusion™ High Fidelity DNA Polymerase	High-fidelity PCR, cloning
										Phusion™ High Fidelity PCR Master Mix with HF Buffer	
										Phusion™ High Fidelity PCR Master Mix with GC Buffer	
										Phusion™ Hot Start Flex DNA Polymerase	
										Phusion™ Hot Start Flex 2X Master Mix	
<i>Taq</i> DNA Polymerase	—	1x (1.3-1.8x10 ⁻⁴) ^b	Yes	—	Yes	No	Yes	Yes	3'A	<i>Taq</i> DNA Polymerase with Thermopol Buffer	Routine PCR
										Hot Start <i>Taq</i> DNA Polymerase	
										<i>Taq</i> 2X Master Mix	
										Hot Start <i>Taq</i> 2X Master Mix	
										Quick-Load <i>Taq</i> 2X Master Mix	
										Multiplex PCR 5X Master Mix	Multiplex end point PCR

Key Take-Home Messages

Central take-home message

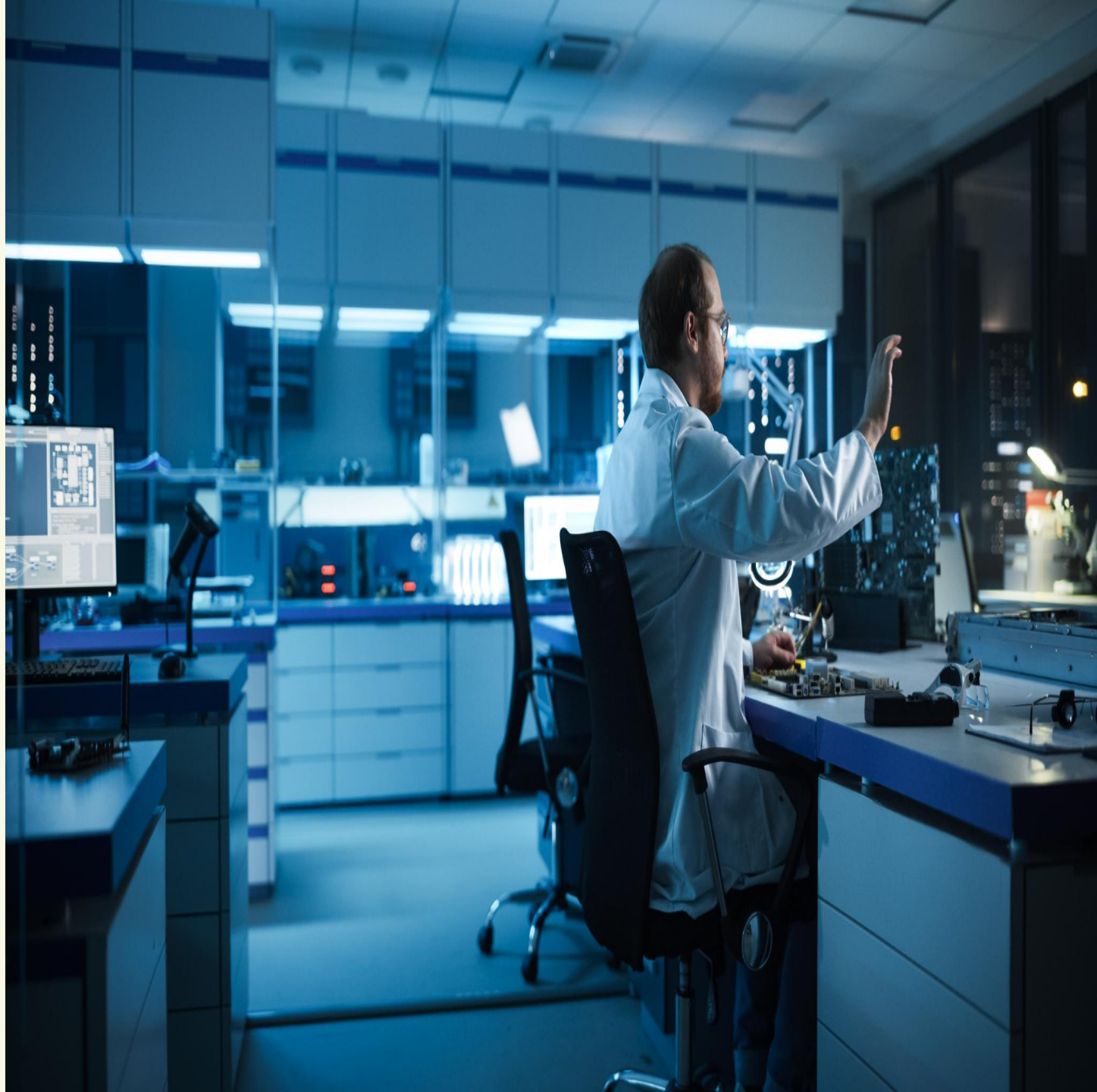
Careful optimization can turn a fragile assay into a robust screening tool. Optimized MAMA-PCR reliably detects single-nucleotide mutations, reduces dependence on sequencing, and accelerates metabolic engineering workflows.

Discussion questions

How broadly will performance generalize across organisms and genomic contexts? Can it be scaled or multiplexed? How does it compare with ever-cheaper sequencing for similar use cases?

Why it matters

The study shows how incremental method improvements can yield outsized gains in efficiency and innovation—and suggests similar optimization principles may strengthen other PCR-based assays.



THANK YOU AND WHAT QUESTIONS DO YOU HAVE?
